RAYAN IBBAHATENE

3EA Student @ ENSEEIHT | Electronics, Electrical Energy & Automatic Control

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PROFILE

Engineering student at **ENSEEIHT** in the **3EA** department following CPGE MP at Lycée Moulay Youssef in Rabat. Passionate about electronics, embedded systems, and software engineering. I aim to **deepen my knowledge** in continuous-system simulations and technical innovation through hands-on engineering challenges.

KEY ENGINEERING SKILLS

Rigorous Analysis: First-principles mathematical modeling & structured problem breakdown.

Complex Problem Solving: Designing robust algorithms and embedded system solutions.

Adaptability & Teamwork: Agility in technical environments & collaborative project delivery.

TECHNICAL KNOW-HOW

Programming & Data

C | Python | MATLAB | SQL | LaTeX

Engineering Domains

Embedded Systems | Automatic Control | Electrical Energy | Linear Algebra | Fluid Dynamics

LANGUAGES

French: Fluent

English: Advanced (Technical)

Arabic: Native

SPORTS & INTERESTS

Football: Teamwork, strategy, and endurance.
Swimming: Discipline and perseverance.
Biking: Commitment and physical stamina.

EDUCATION

Engineering Degree Program (3EA)
2026 – Present

ENSEEIHT (n7) — Toulouse, France

Advanced curriculum specializing in Electronics, Electrical Energy, Systems, and Automatic Control.

CPGE MP (Mathematics & Physics)
2024 – 2026

Lycée Moulay Youssef — Rabat, Morocco

Intensive 2-year preparation in theoretical mathematics, applied physics, and fundamental computer science.

Baccalauréat in Mathematical Sciences
2024

Rabat, Morocco

Rigorous national curriculum focused on advanced mathematical sciences and physics.

ENGINEERING PROJECTS

TIPE Project — Topological Analysis of Vascular Networks
2025 – 2026

Research Theme: "Cycles et Boucles" | Python, Linear Algebra, Graph Theory

- **Modeling:** Studied topological robustness of vascular transport networks under localized damage.
- **Matrix Analysis:** Formulated Laplacian matrices and computed Fiedler algebraic connectivity vectors to evaluate loop structural resilience.
- **Results:** Demonstrated the key role of closed-loop topographies in preserving hydraulic transport stability.

Terminal-Based Snake Game Engine
Personal Project

C Language, Data Structures, Systems Programming

- **Conception:** Built a classic terminal-based Snake game engine from scratch using pure C.
- **Implementation:** Designed a dynamic memory-managed game board state with custom multi-dimensional array layouts.
- **Architecture:** Structured modular code base demonstrating low-level memory management and direct input handling.